



ESCUELA DE
NEGOCIOS

DIVISIÓN ACADÉMICA DE
ADMINISTRACIÓN Y CONTADURÍA

CORPORATE FINANCE

LICENCIATURA EN DIRECCIÓN FINANCIERA

COURSE DESCRIPTION

This course covers the foundations of finance with an emphasis on applications that are vital for corporate managers. We will discuss many of the major financial decisions made by corporate managers both within the firm and in their interactions with investors. Essential in these decisions is the process of *valuation*, which will be an important emphasis of the course. Topics include criteria for making investment decisions, valuation of financial assets and liabilities, relationships between risk and return, capital structure choice, the effective use and valuation of derivative securities, as well as, risk management, and real options.

REQUIRED MATERIALS AND SUPPLEMENTARY READINGS

Required Materials:

1. Textbook. Readings on substantive matters are available in:

***Corporate Finance (plus MyFinanceLab)*, Jonathan Berk and Peter DeMarzo, 3rd ed., Pearson - Prentice Hall, 2013.**

The Berk-DeMarzo (plus MyFinanceLab) textbook is the **only required textbook**. You are expected to read the relevant book chapters **before** each class (please refer to the **Course Outline**, below for details). When buying the book, **make sure it includes access to MyFinanceLab**, the online platform for homework assignments.

2. Cases:

A list of cases and their due dates is described in detail in the course outline (see below).

3. Lecture notes:

There will be **lecture notes** for each lecture. These materials will be available electronically.

Supplementary Readings (not required):

For those interested in an introductory guide to financial terms and markets please refer to:

Guide to Money and Investing, by Virginia B. Morris and Kenneth Morris, Lightbulb Press, 2015 (www.lightbulbpress.com), or any prior edition.

For those interested in some entertaining supplementary reading on financial markets, the following books can be purchased in many bookstores or from online vendors:

Capital Ideas: The Improbable Origins of Modern Wall Street, by Peter L. Bernstein, Wiley, 2009, or any prior edition.

A Random Walk Down Wall Street, by Burton G. Malkiel, W. W. Norton & Company, 2015 or any prior edition.

Reading *The Wall Street Journal*, *The Financial Times*, *El Financiero*, *El Economista*, *The Economist*, or the business section of a major newspaper, is fundamental for any student interested in finance. Class discussion will often reference news items.

CLASS PREPARATION

There will generally be **both** a **reading assignment** and a **case** or other problems to prepare for each session. You should prepare each class by: (a) reading the relevant material from Berk-DeMarzo, (b) reading the **case**, and (c) working on the **assigned case or problems**. Students are encouraged to discuss the cases and other content of the course with other students.

Cold-calling and in-class quizzes may be used to check how well students are prepared. Note that the assignments listed in the syllabus are subject to change during the quarter; changes will be announced in class when applicable.

The course covers a large amount of material in significant depth. You are strongly encouraged to be prepared for class and to **keep up with the material as we go along**, since it may be difficult to catch up if significant lags in preparation occur.

Class participation is critical both for your own learning and that of your colleagues in the course. Participation, which includes attendance and in-class quizzes, will count for part of the grade (see below). Absence from class will adversely affect this part of your final grade.

This course is structured so that each lecture's material builds upon the concepts introduced in prior lectures. *It is therefore **crucial** that you stay up to date with the course readings and assignments. It will be difficult to catch up once you have fallen behind.* To reinforce and test your knowledge, **homework assignments in MyFinanceLab** will be provided (see outline).

GRADING

Your course grade will be determined by your class participation and in-class quizzes (15%), performance on each of the two midterm exams (30%), the final exam (40-55%), and completion of recommended homework assignments available through MyFinanceLab (0-15%).

The purpose of the weekly MyFinanceLab homework is to keep you up to date with the course material. The amount you complete correctly will determine the weight put on homework in your final grade, up to a maximum of 15%. The remaining weight will be put on the final exam. (For example, if 1500 total homework points are possible, and you complete correctly 1000, homework will count for 10% of your grade, and the final will count for 45%). Because students have unlimited attempts to get assignments right (*before each assignment is due*), almost all students receive full points on the homework. This means that not doing an assignment puts you at a considerable disadvantage relative to others when the final grades are computed.

ATTENDANCE POLICIES

It is expected that all students: (a) attend all classes, (b) will be on time for class, and (c) will not disrupt the class by leaving early or by leaving and returning to class. I recognize that there are medical and other special circumstances that may cause a student to miss a class or part of a class. If such circumstances arise, students **are required to email me** (fpg@itam.mx).

Please explain the reason for the absence in the body of the message. This should be done before the absence if it is foreseen or as soon as possible after the absence if it was unanticipated. Students who miss parts or entire class sessions without reasonable justification should expect that their class participation grade will be adversely affected in a significant way.

COURSE ETIQUETTE

Laptops, Tablets, Smartphones, etc. may be used in class as long as they are used for note-taking or to support class participation. Hence, open laptops will be taken as an indication of students' willingness to participate in class discussion. Other electronic devices should only be used accordingly: (a) for note-taking purposes, or (b) to support direct class participation.

EXAMS AND MAKEUP CLASSES

The **midterm exams** will be **in-class** given: (1) **Tuesday, February 16th, 2016**, and (2) **Tuesday, April 19th, 2016**. The **final examination** date and time will be determined by Servicios Escolares. There will be no scheduled alternatives. There will be **two make-up classes scheduled**: (1) Friday, February 5th and (2) Friday April 29th. To accommodate everyone, these make-up classes will be taught both from 7am to 10am, and from 6pm to 9pm on both dates.

Class	Date	Day	Topic	Cases and Readings	Readings †
1			Time Value of Money, Present and Future Values, NPV <u>Key ideas:</u> financial statements: basics; time value of money net present value (NPV), and annuity formulas		Chp. 4: Time Value of Money (§4.1-4.6: pp. 96-121)
2			Financial Decision-Making: Free Cash Flow <u>Key ideas:</u> NPV ingredients: (a) cash-flows and (b) discount rates Free cash flows (FCFs): earnings are not FCFs Discount rates: (a) time value and (b) compensation for risk	Callaway Golf FX-1, Part A	Chp. 8: Capital Budgeting (§8.1-8.2: pp. 234-246)
3			Scenario & Sensitivity Analysis; Application <u>Key ideas:</u> Computing FCFs: scenario and sensitivity analysis Annuity formulas and annuity spreadsheet Effective annual rates (EAR), annual percentage rates (APR)	Callaway Golf FX-1, Part B Should you pay cash?	Chp. 8: Capital Budgeting (§8.5: pp. 252-257) Chp. 4: computing loan payments: pp. 122-125 Chp. 5: Interest Rates (§5.1-5.2: pp. 142-147)
4			Capital Budgeting: Incremental Cash Flows <u>Key ideas:</u> Computing FCFs: incremental cash-flows, timing of CFs, terminal values, sensitivity analysis The link between risk and the cost of capital	Diamond Chemical, Part A	Chp. 8: Capital Budgeting (§8.5: pp. 252-257)
5			Capital budgeting: investment decision rules <u>Key ideas:</u> NPV and the Internal Rate of Return (IRR) IRR as bank savings on a real investment. The limitations of using the IRR rule. Project selection with resource constraints	Diamond Chemical, Part B	Chp. 7: Investment Decision Rules (§7.1-7.5: pp. 206-224)
6			Interest Rates and Bond Valuation: Basics <u>Key ideas:</u> Term structure of interest rates, the yield curve Bond basic characteristics: face value, coupons, maturity, STRIPS, yield to maturity, zero-coupon bonds, etc Evolution of bond prices over time		Chp. 5: Interest Rates (§5.3: pp. 147-153) Chp. 6: Valuing Bonds (§6.1-6.2: pp. 169-180)
7 and 8			Bond Pricing and No-Arbitrage <u>Key ideas:</u> pricing bonds using no-arbitrage; Interest rate sensitivity of bonds MAKEUP CLASS Firm Valuation I <u>Key ideas:</u> Equity cost of capital, capital gains, dividend yield Valuation models: (a) dividends, (b) constant growth model (c) total payout model	Orange County	Chp. 6: Valuing Bonds (§6.3: pp. 180-184) Chp. 9: Valuing Stocks (§9.1-9.3: pp. 271-283)
9			Firm Valuation II <u>Key ideas:</u> (a) Free cash flow valuation models (b) valuation based on comparable firms ("multiples")		Chp. 9: Valuing Stocks (§9.3-9.4: pp. 284-292)
10			Valuing Interco <u>Key ideas:</u> different approaches to valuing a firm Assumptions behind alternative valuation approaches Main: (a) discounted cashflow and (b) multiples	Interco Interco data	
11			Midterm exam 1		
12			Risk, the Cost of Capital, Portfolio Choice, Diversification <u>Key ideas:</u> Expected returns, variance and standard deviation Risk-return tradeoff, covariance, correlation across assets Portfolio risk and diversification		Chp. 10: Capital Markets and the Pricing of Risk (§10.1-10.6: pp. 312-335)
13			Describing portfolios and Optimal Efficient Frontier <u>Key ideas:</u> Optimal investment frontier. Optimal portfolios with short-selling. Portfolios with risk-free investments and leverage Optimal portfolios; Sharpe ratios; Minimum required rate of return	Stanford Management Co. Part A Stanford Mgmt data	Chp. 11: Optimal Portfolio Choice and the Capital Asset Pricing Model (§11.1-11.5: pp. 351-375) In particular: formulas & example in pp. 354-359

Class	Date	Day	Topic	Cases and Readings	Readings †
14			Market Pricing of Risk <u>Key ideas:</u> Capital Asset Pricing Model (CAPM). CAPM: the market is the tangent portfolio. Capital market line: max expected return per unit of volatility. Security market line: market portfolio as benchmark for other investments. Beta of: (1) a security, (2) an asset are not the same. Using betas to compute the expected return of a security	Stanford Management Co. Part B Portfolio Optimizer	Chp. 10: Capital Markets and the Pricing of Risk (\$10.7-10.8: pp. 337-342) Chp. 11: Optimal Portfolio Choice and the Capital Asset Pricing Model (\$11.6-11.7: pp. 375-380)
15			Estimating the Cost of Capital in Practice <u>Key ideas:</u> CAPM implementation: risk-free, market index, betas and risk-premia; unlevered betas; equity & debt betas. Estimating betas using historical returns; Using comparables in estimating betas and the cost of capital; formulas to unlever and re-lever betas	Cost of Capital at Ameritrade Ameritrade data	Chp. 11: Optimal Portfolio Choice and the Capital Asset Pricing Model (\$11.7: pp. 379-386) Chp. 12: Estimating the Cost of Capital (\$12.1-12.7: pp. 400-423) Chp. 14: Levered and Unlevered Betas (pp. 492-494)
16			Valuing a Firm <u>Key ideas:</u> Valuing a firm in practice: (a) project cashflows, and (b) estimate discount rates. Projecting cash-flows: link between forecasts & operations. Estimating the cost of capital: CAPM; Timing of cash-flows and terminal values	Clarkson Lumber Clarkson data	
17			Financing Investment: Perfect Capital Markets <u>Key ideas:</u> differences between debt and equity. Debt is senior (less risky) relative to equity: a lower cost does not imply that debt is "cheaper." Difference between unlevered and levered equity Weighted average cost of capital (WACC) & project-specific risk Modigliani-Miller propositions: (a) debt does not affect value (b) WACC does not change with leverage		Chp. 14: Capital Structure in a Perfect Market (\$14.1-14.3: pp. 478-494)
18			Financing Investment: Fallacies and taxes <u>Key ideas:</u> Capital structure fallacies: (a) dilution, (b) earnings per share. Beyond MM propositions. Corporate income taxes & value: interest tax shields. With corporate taxes, leverage affects the WACC and firm value. Who captures the interest tax shields?		Chp. 14: Capital Structure in a Perfect Market (\$14.4: pp. 495-499) Chp. 15: Debt and Taxes (\$15.1-15.3: 508-519)
19			Financing Investment: Methods to Value the Leverage Effects <u>Key ideas:</u> Main methods to evaluate the effect of leverage on firm value: (a) WACC and (b) Adjusted present value (APV). Explore why firms are not 100 percent debt financed. Extensions to personal taxes		Chp. 18: Valuation with Leverage (\$18.1-18.3: pp. 628-636)
20			Valuation with Leverage: WACC <u>Key ideas:</u> The impact of the costs of bankruptcy and financial distress (FD) on financing decisions; Picking a target capital structure; Weighted Average Cost of Capital (WACC); Valuing a going concern; practical Issues in valuation	Mercury Athletic Mercury Athletic data	Chp. 18: Valuation with Leverage (\$18.1-18.3: pp. 628-636) Chp. 16: Financial Distress and Incentives (\$16.1-16.7: pp. 539-564)
21			Agency Costs of debt and Adjusted Present Value <u>Key ideas:</u> Agency costs of debt: (a) excessive risk-taking, (b) debt overhang. Agency benefits of leverage. Debt can act as a disciplinary device: force managers to payout. Expanded tradeoff theory: taxes, FD, agency and incentives. WACC and APV valuation	Sealed Air	Chp. 16: Financial Distress and Incentives (\$16.1-16.7: pp. 510-533)

Class	Date	Day	Topic	Cases and Readings	Readings †
22			Leverage: The Role of Information <u>Key ideas:</u> Capital structure decisions as signals of firm quality The role of information in capital structure decisions & firm value Pecking order hypothesis. Implications of information asymmetry for equity issuance.		Chp. 16: Distress, Incentives and Information (\$16.8: pp. 564-572)
23			Capital Structure: Summary and Applications <u>Key ideas:</u> Lessons from the recent financial crisis. Discussion on the capital structure of banks, tax policy. Payout policy: dividends and share repurchases		
24			Risk Management: When does RM add value? <u>Key ideas:</u> Rationale for risk-management. Hedging as insurance Forward: contract to buy/sell in the future at a price set today Future contracts: forward contract that is traded on an exchange		Chp. 20: Financial Options (\$20.1-20.2: pp. 707-718) Chp. 30: Risk Management (\$30.1: pp. 986-992)
25			Midterm exam 2		
26			Financial Options I <u>Key ideas:</u> Options: basic characteristics and payoffs. Call (put) options: right but not the obligation to buy (sell) an asset. Using options for risk-management purposes. Options vs. forwards in risk-management		Chp. 20: Financial Options (\$20.1-20.2: pp. 707-718)
27			Financial Options II <u>Key ideas:</u> Put-call parity Binomial option pricing: call options		Chp. 20: Financial Options (\$20.3-20.5: pp. 718-726) Chp. 21: Option Valuation (\$21.1-21.2: pp. 739-746)
28			Financial Options III <u>Key ideas:</u> Binomial option pricing: multi-period binomial option pricing: put options. Black-Scholes formula		Chp. 20: Financial Options (\$20.3-20.5: pp. 718-726) Chp. 21: Option Valuation (\$21.1-21.2: pp. 739-746)
29 and 30			MAKEUP CLASS Determinants of Option Values <u>Key ideas:</u> Using the Black-Scholes formula in practice Black-Scholes calculator in excel. Valuation of employee stock options (ESOs) Challenges of using Black-Scholes formula for valuing ESOs Implied volatility; Determinants of option prices Dynamic Decision-Making and Real Options <u>Key ideas:</u> Real options: flexible investment projects are like financial options: (a) option to wait, (b) option to expand, (c) option to delay. Implications for investment in practice. Compare a DCF valuation to a real options valuation	Retaining Top Talent Options Calculator	Chp. 21: Employee stock options: pp. 756 Chp. 20: Determinants of option prices: pp. 720-722 Chp. 21: Implied volatility: pp. 752-753
				Deciding When to Build	Chp. 22: Real Options (\$22.1-22.4: pp. 774-789)
31			The value of staging investments <u>Key ideas:</u> the value of staging investments in the pharma and venture capital industries: (a) set milestones in advance (b) postpone substantial investments until information arrives (c) drop projects that do not meet the milestones		Chp. 22: Real Options (\$22.1-22.4: pp. 774-789)
32			Corporate Securities as Options and Credit Risk <u>Key ideas:</u> equity can be interpreted as a call option on firm value Equity holders right to default is like a put option on the firm's assets Using Black-Scholes to compute the value of debt and equity No class		Chp. 20: Options and Corporate Finance (\$20.6: pp. 727-731) Chp. 21: Corporate Applications of Option Pricing (\$21.5: pp. 763-767)
33			Course Overview		